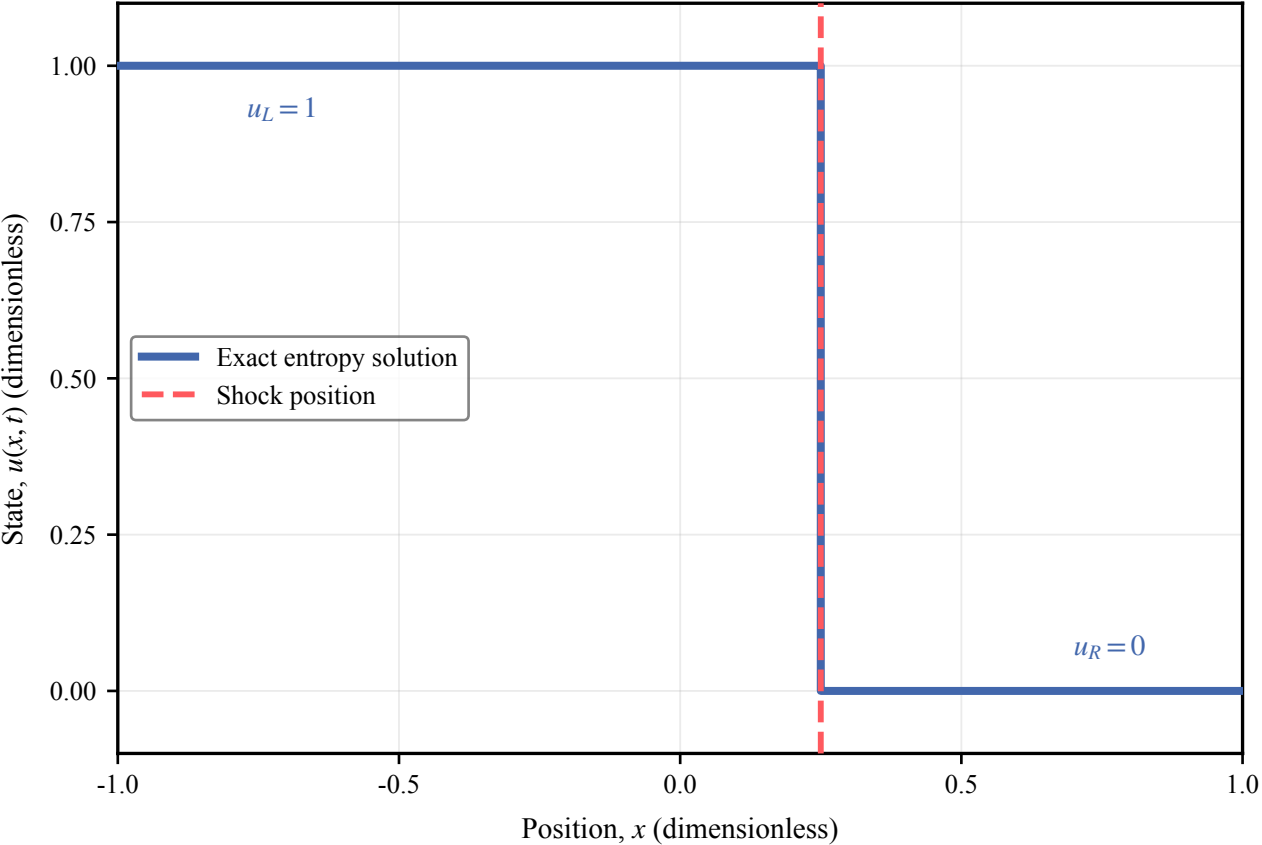


One-Dimensional Inviscid Burgers Equation

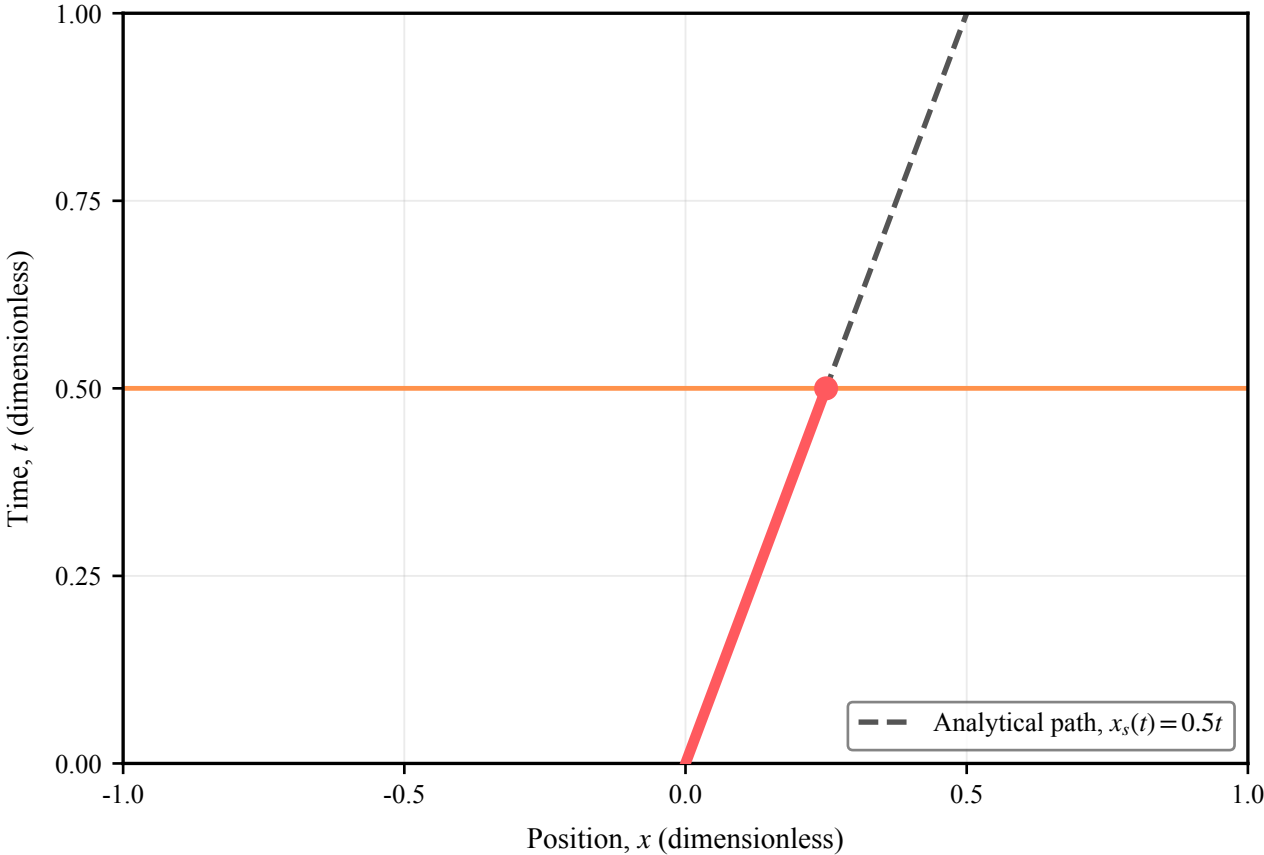
$$u_t + (u^2/2)_x = 0, \quad u_L = 1, \quad u_R = 0, \quad s = 0.5$$

Entropy-Solution Profile

$$t = 0.50, \quad x_s(t) = 0.25$$



Shock Trajectory in Space-Time



The discontinuity moves at constant speed $s = 0.5$; the exact inviscid shock does not spread.